

Git, Part 2: The Humans

User Study Day · Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

The Humans

Open the class link (it ends in /thu) and type your first name.

Do Developers Really Know How to Use Git Commands? A Large-Scale Study Using Stack Overflow

Yang, Zhang, Pan, Xu, Zhou, Huang · ACM TOSEM 2022

Take the paper's survey

The real questions, from the authors' published form.

92 developers answered them. You are next.

2. What is your professional area?
3. What is your highest degree?
4. How many years have you been using Git?
5. Which of the following levels do you consider your ability to use Git to be?
6. What are the main approaches you used to learn Git commands?
7. Do you have any suggestions for potential new developers on learning Git commands

Q1 of the form, the donation choice, is left out.

1 · THE QUESTION

The paper in one slide

RQ1 · How many Git questions?	80,370 questions, 2008–2020. Since 2010, about 0.4% of all Stack Overflow questions each year.
RQ2 · Who asks?	In 2020, 40.0% of Git askers had registered more than 5 years earlier (21.2% of all askers).
RQ3 · Which commands?	Most viewed on average, among commands in 200+ questions: <code>git revert</code> , <code>git reflog</code> , <code>git stash</code> , <code>git clean</code> , <code>git reset</code> .
RQ4 · Which are hardest?	Most often without an accepted answer: rare commands; among frequent ones, <code>git credential</code> and <code>git submodule</code> .
RQ5 · How do people learn?	Survey, 92 of 508 invited: 81.7% of ticked approaches are self-learning.

What kind of study is this?

Need-finding: what problems do people have?

Formative: which solution looks promising?

Evaluative: did our tool work?

1 · THE QUESTION

Need-finding

Need-finding: what problems do people have? ✓

Formative: which solution looks promising?

Evaluative: did our tool work?

Its shape: what problems do developers face when using Git commands?

A need-finding study must see what goes badly **and** what goes well. Stack Overflow questions record what people got stuck on; when Git just worked, nobody asked.

Is the question interesting?

Stack Overflow posts are...

Observation: watching people use Git

Traces: records left by real work

Self-report: people telling us

2 · THE DATA

Traces. Of asking.

Observation: watching people use Git

Traces: records left by real work ✓

Self-report: people telling us

Stack Overflow questions are on Sarah's list of traces: better than asking people, worse than watching them.

They record people asking, not people using Git. A common problem is asked once, then viewed millions of times.

RQ5 is self-report. Nobody in this paper was watched using Git.

Can this data answer "do developers know how to use Git commands?"

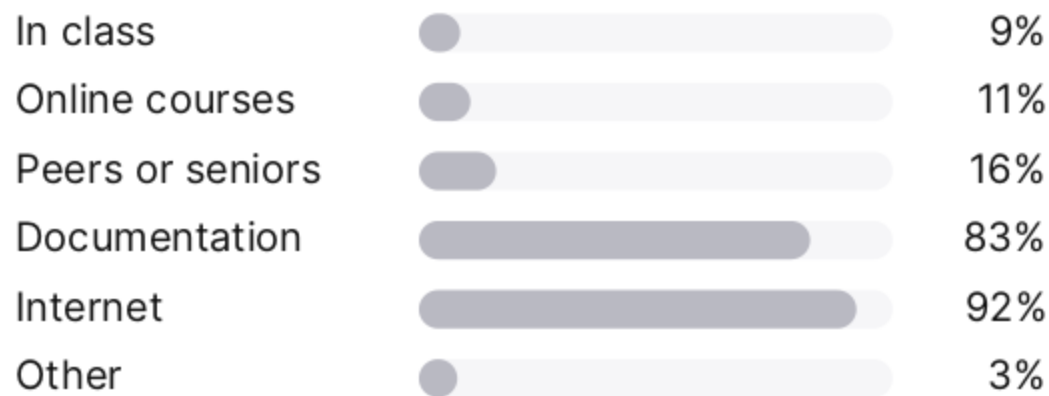
2 · THE DATA

You and the paper's 92

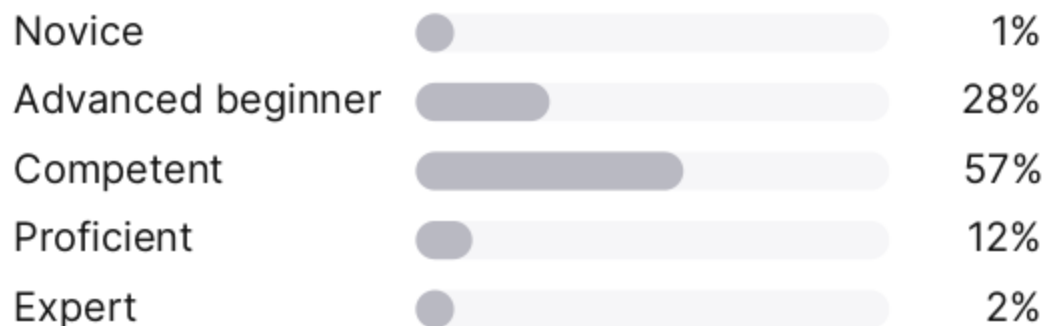
81.7% is 161 of 197 ticked boxes. By people: 85 of 92 ticked the internet.

Invited: developers who had recently asked a Git question on Stack Overflow, plus some researchers. 92 of 508 answered.

THE PAPER'S 92: HOW THEY LEARNED (SHARE OF PEOPLE)



THE PAPER'S 92: GIT LEVEL, SELF-RATED



Median years using Git: 8. In class, the app shows your answers next to these.

Which of these numbers would you trust? Did Tuesday change your answer?

Check the rule on 8 real posts

The paper's rule: a post counts for every Git command in its question **or its accepted answer**.

For each post: is it really a question about that command?

On your laptop: 8 posts, each with the question, the accepted answer, and the command the paper counted it for.

Randomly sampled from the paper's own data (analysis/thursday_numbers.py, seed 294).

3 · THE ANALYSIS

The rule and you

The rule counts all 8 posts. In 5, the command appears only in the answer.

Our re-run on all 80,370 posts: the command is only in the answer for 55% of `git reflog` posts, 44% of `git revert`, 44% of `git reset`.

POST	COUNTED FOR	ASKER NAMES IT
git how to move back commit (6,163 views)	git reflog	Yes
How do I remove a single file from the staging area (undo git add)? (656,913 views)	git reset	Answer only
Undo a Git merge that hasn't been pushed yet (2,371,183 views)	git reflog	Answer only
How can i undo changes between two commits? (24 views)	git revert	Answer only
git 'merge' only existing files (1,378 views)	git clean	Yes
Git branch not working the way I expected it to (2,989 views)	git reset	Yes
revert a commit after push, checkout a commit behind and modification of files (46 views)	git revert	Answer only
Revert erroneous git checkout (45 views)	git stash	Answer only

Did they measure “developers find this command hard”?

3 · THE ANALYSIS

One question, 9.1 million views

“How do I undo the most recent local commits in Git?” names no command. Its accepted answer counts it for five. Without it, `git reflog` falls from #2 to #8.

	PAPER: MEAN VIEWS	MEDIAN VIEWS	ASKER NAMES IT
<code>git revert</code>	#1	#60	#1
<code>git reflog</code>	#2	#57	#40
<code>git stash</code>	#3	#39	#6
<code>git clean</code>	#4	#27	#42
<code>git reset</code>	#5	#43	#4

Our re-run of the paper's rule on its data. Rank among the 62 commands in 200+ questions, the paper's filter.

Do you believe their answer to RQ3?

BREAK

Break

5 minutes. Then groups.

Put one claim on trial

Your group gets one of the paper's claims.

Write what they measured, then the strongest sentence the data supports.

RQ2 ANSWER, §3.2

"This suggests that even developers with years of development experience can have trouble using Git commands."

§1, FINDING (4); RQ4, §3.4

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

ABSTRACT; RQ5, §3.5

"Self-learning is the primary learning approach."

CONCLUSION, §7

"the number of questions related to Git commands has been growing steadily"

4 · DO YOU BELIEVE IT?

The verdicts (1/2)

The title asks whether developers know how to use Git commands. Nothing in the paper watches anyone use one.

"This suggests that even developers with years of development experience can have trouble using Git commands."

THEY MEASURED

Years since the asker registered on Stack Overflow.

THE DATA SUPPORTS

Many Git questions come from Stack Overflow accounts that are years old.

"...for the more frequently-used commands, git credential and git submodule are among the most difficult ones."

THEY MEASURED

The share of a command's questions where the asker never marked an answer as accepted.

THE DATA SUPPORTS

These questions end without an accepted answer more often (43.0%, 37.5%) than Git questions overall (36.5%, our re-run).

Which claim survives best?

4 · DO YOU BELIEVE IT?

The verdicts (2/2)

The title asks whether developers know how to use Git commands. Nothing in the paper watches anyone use one.

"Self-learning is the primary learning approach."

THEY MEASURED

What 92 of 508 invited people ticked. The practitioners invited had recently asked a Git question on Stack Overflow.

THE DATA SUPPORTS

Most of these 92 say they learned from the internet (85) and the documentation (76).

"the number of questions related to Git commands has been growing steadily"

THEY MEASURED

Git questions per year (Fig. 2a), and their share of all Stack Overflow questions (Table 1).

THE DATA SUPPORTS

Questions peaked in 2016 (9,125), fell to 7,076 in 2019, then rose again. Their share has held at about 0.4% since 2010.

Which claim survives best?

Design for the need that survived

The need: undo. The most-viewed Git question asks how to undo the most recent local commits.

Its accepted answer's fix is `git reset HEAD~`. Its further reading points to `git reflog`: the record Git already keeps of where HEAD has been.

ONE CHANGE TO GIT

e.g. what the user sees or types

THE STUDY THAT WOULD SHOW IT WORKS

Does ... increase/decrease ... for ...
doing ...?

YOUR DATA

Watch people, collect traces, or ask them? Which, exactly?

Your designs

Each group reads its change, the study that would show it works, and its data. The app shows all of them.

Which of these could you test by watching people?

What Git and Jujutsu did

Git 2.23 (2019) split `git checkout` into `git switch` and `git restore`. In 2020 a developer “working with Git for years” asked what `git checkout [file]` does next to `git restore`: the paper’s reference [41].

Jujutsu (`jj`) logs every operation, and `jj undo` reverses the last one.

Which data would convince Git’s maintainers your change works?

EXIT

2 min

One new idea for building tools for people

One line, on your own.

THURSDAY

Asking is not using.

Tuesday: flat history is data loss. Thursday: asking is not using.

Both: check what your data actually records before you believe it.